

PAPER_200: Um Universal Magnetism Taxonomy — Complete Variant Catalogue

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Session: 50 — grokshare7514fe.txt Full Audit

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$$F_U(r,t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$L_{\text{UQFF}} = \frac{4\pi G M c}{\kappa_{\text{es}}} \text{Bigl}(1 - [SSq] \cdot e^{-\kappa \Delta t} \text{Bigr}), \quad [SSq] = 0.57$$

<!-- ? = 5.0e-4 day¹, [SSq] = 0.57, $\beta_i = 6.1e-1$ -->

Abstract

Universal Magnetism (Um) in UQFF is a general magnetic energy density operator that scales with vacuum density ρ_{vac} and an exponential damping factor $(1-e^{-\tau t})$. This paper catalogs all named Um sub-variants extracted from the BBCEquations04Sept2025.pdf, covering 50+ unique astrophysical and cosmological regimes: cosmological magnetism, reheating, BBN, MHD dynamo, quasi-normal modes, Blandford-Znajek, photo-evaporation, planet migration, terminal velocity, Press-Schechter, star formation efficiency, BH entropy, evaporation, angular momentum, jet velocity, GW chirp mass, QNM, duty cycle, peculiar velocity, ionization, deuterium, Friedmann-1, QNM damping, Arnett, reheating, Kazantsev dynamo, Alfvén Mach, field reversal, and more.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Universal Magnetism General Form

$$U_{m,X}(\arg) = [S_j \mu_j(t, \rho_{vac}, [SCm])/r_j] \times (1 - e^{-\tau t} \cdot \cos(\pi t_n)) \times F_X$$

or (single j variant):

$$U_{m,X}(\arg) = \mu(\rho_{vac}) \times (1 - e^{-\tau t}) \times F_X$$

where:

μ_j = magnetic moment density at scale j

$\rho_{vac}, [SCm]$ = superconductive vacuum density

τ = exponential decay rate

t_n = normalized time: $t_n = t/t_{\text{Hubble}} \cdot (1+H(z) \cdot t_0)$

F_X = system-specific magnetic function

2. Cosmological Magnetism Variants

Symbol	F_X Expression	Physical Context
Um,cosmo(t,a)	$(k_c^2/a^2)/H^2$	UQFF cosmological magnetic field scaling
Um,reh(N)	$(30V_{end}/(p^2 g^*))^{1/4} \cdot e^{-3N_{reh}/4}$	Reheating post-inflation
Um,BBN(t)	$v(3/(32pG?_{rad})) \sim 180 \text{ s } (T \sim 0.1 \text{ MeV})$	Big Bang Nucleosynthesis
Um,fried1(a)	$Om(1+z)^3 + Ok(1+z)^2 + O_{-?}$	Friedmann-1 Hubble term
Um,wde(a)	$? ? a^{-3(1+w)}$	Dark energy equation of state
Um,deden(a)	$w(a) = w_0 + w_a(1-a)$ (CPL form)	Dark energy density evolution
Um,grow(a)	Growing mode $D ? a$ (matter era), suppressed by DE	Structure growth factor
Um,curv(?)	$?? = v(2e) \cdot H \cdot M_{pl}$	Curvature perturbation from inflation
Um,ng(f)	From $d? curvature$ on superhorizon scales	Non-Gaussianity

3. Black Hole Thermodynamics Variants

Symbol	F_X Expression	Physical Context
Um,haw(M)	$T = ?/(2p) ; ? = c4/(4GM)$	Hawking temperature surface gravity
Um,bhent(M)	Holographic $S ? A/l_{pl}^2$	Bekenstein-Hawking entropy area law
Um,evapl(M)	$dM/dt = -P/c^2$	Black hole evaporation mass loss rate
Um,qnm(a)	$a \cdot c^3/(2GM) \times [l=2, m=2]$	QNM ringdown frequency (Berti fits)
Um,damp(a)	$Q \sim 10$ for $l=2, m=2$	QNM quality factor (damping)
Um,bz(a)	Power $? B^2 O^2 H \cdot R4H$ (electromagnetic extraction)	Blandford-Znajek energy extraction
Um,bz2(a)	$?(16p)F^2BH \cdot O^2BH/c$	BZ jet power (EHT-updated form)

4. Compact Object and Stellar Variants

Symbol	F_X Expression	Physical Context
Um,MHD	$4p? \cdot v^2 turb \cdot MA$	MHD turbulent Alfvén scaling
Um,termv(G)	$L/LEdd \sim 1 ; v8 = v(2(L/?c) \cdot v^2_{esc})$	Terminal velocity in radiation winds
Um,sfe(z)	$?_* ? M^{1.1} \cdot (1+z)^{2.25}$	Star formation efficiency evolution
Um,duty(t)	$\exp(-t/t_{cool})$	AGN feedback duty cycle
Um,ang(t,r)	$v(GM/r^3) \cdot r_A ; accretion \text{ adds } L=?r^2O$	Angular momentum accretion
Um,jetvel(r)	$O \cdot rA \cdot (rA/r0)^{1/2}$	Jet velocity from Alfvén radius
Um,glitch(t)	Angular momentum transfer $?L = I_s \cdot ?O$	Superfluid vortex glitch
Um,ms(M)	Eddington: $L \sim \mu 4M^3$ (opacity, composition μ)	Main-sequence L-M relation
Um,ml(M)	Calibrate from HR diagram	Mass-luminosity relation
Um,relax(?)	$s_v^3/(G^2?m)$	Two-body relaxation time
Um,evap(N)	Integrate exponential cluster dissolution	Star cluster evaporation
Um,star(N)	$136 \cdot t_{relax}/\ln(0.02N)$	N-body cluster relaxation timescale

5. Gravitational Wave Variants

Symbol	F_X Expression	Physical Context
Um,chirp(f)	$(32/5)(G\mu^5/c^5)(pf)^{10/3}$	GW chirp mass inspiral energy loss
Um,orbdec(e)	$dE/dt = -L, E = -GM\cdot\mu/(2a)$	Orbital decay by GW quadrupole radiation
Um,peri(a)	Kepler: $a^3/P^2 = GM/(4p^2)$	Post-Keplerian periastron advance
Um,kilo(t)	Diffusion $t_d^2 = 3?M/(4pcv^2)$	Kilonova light curve peak

6. ISM and Plasma Variants

Symbol	F_X Expression	Physical Context
Um,malf(B)	$M_A < 1$ sub-Alfvénic (kinetic $\sim B^2/(8p)$)	Alfvén Mach number turbulence regime
Um,rev(?)	$?B/?t = ?\times(aB - ??\times B)$ (growth vs diffusion)	Magnetic field reversal (parity)
Um,dyn(k)	$?E dk ; E_M = ?E\cdot dk$ (Kazantsev E spectrum)	Magnetic energy growth via dynamo
Um,alf(B)	Wave speed from linearization $?^2b/?t^2 = v_A^2\cdot ?^2b/?z^2$	Alfvén wave propagation
Um,turb(l)	$?v^2/t_{eddy} ? l^{2/3}$ (Kolmogorov-like ISM)	Turbulent energy cascade rate
Um,cshock(z)	$(?\times B)\times B/(4p) + ?i\cdot ?in\cdot(vn-vi)$	C-type shock (magnetized, continuous)
Um,jshock(M)	$v1/cs \times T2 ? vs^2$	J-type shock (discontinuous) Rankine-Hugoniot
Um,dsa(p)	$B?^1\cdot pc/e \times ? ? B^2$	Diffusive shock acceleration spectrum

7. Cosmological Structure Formation Variants

Symbol	F_X Expression	Physical Context
Um,ps(M)	$s(M)$ rms fluctuation $? d \ln s/d \ln M$	Press-Schechter halo mass function
Um,halo(z)	$?P(k)W^2(kR)d^3k/(2p)^3 \times (1+z)^{m/2}$	Extended Press-Schechter halo mergers
Um,mig(a)	$da/dt = 2G/(Mpv(GM^*\cdot a))$	Type-I planet migration
Um,migr(a)	$S(H/r)?^3$	Disk migration timescale
Um,acc(t)	Accretion $? = L/(ec^2)$	BH accretion (Eddington-limited)
Um,disk(?)	$?* ? M^{3/2}_{gas}/R^{3/2}$	Star formation rate in disk
Um,burst(t)	Enhancement $10-100\times$ at low z	Merger-induced starburst enhancement
Um,metal(Z)	$Z = Y\cdot SFR/?gas - Z\cdot ?out/M_{gas}$	Metal enrichment mass balance
Um,sfe(z)	Efficiency limited by cooling/feedback	Star formation efficiency
Um,fb(?)	$?_*\cdot SFR$ (fraction ?)	Feedback energy injection (SN/AGN)
Um,bhmf(z)	$erfc(dc(z)/v(2)s(M',z)) \times extend to Mf$	BH mass function (EPS)

8. Reionization and Early Universe Variants

Symbol	F_X Expression	Physical Context
Um,ion(t)	$?aB\cdot n^2\cdot e\cdot C\cdot dt$	Ionization fraction integrated

Symbol	F_X Expression	Physical Context
Um,reion(t)	$\int aB \cdot n^2 e \cdot C \cdot dt$ (full Madau model)	Cosmic reionization
Um,eta(t)	Fit networks to D, He abundances	Baryon-to-photon ratio η
Um,deb(T)	Weak freeze at $T \sim 1$ MeV; D photodissociation below	Deuterium bottleneck onset
Um,recomb(z)	Optical depth $\tau = \int n_e \cdot n_p \cdot dl$	Recombination epoch
Um,cmb(k)	Transfer $\mathcal{T}_l$: Sachs-Wolfe + acoustic oscillations	CMB power spectrum
Um,bub(t)	Expand for moving ionization front	Bubble growth during reionization

9. Dark Matter and Cluster Variants

Symbol	F_X Expression	Physical Context
Um,nfw(r)	Fit to universal NFW form $\rho_s/(x \cdot (1+x)^2)$	NFW dark matter halo profile
Um,nfwrot(x)	Flat rotation curve for $r \gg r_s$	NFW rotation curve
Um,sidm(t)	Exponential density flattening ($G \cdot t \sim 1$)	SIDM core formation
Um,vir(r)	$s^2_v = GM/(3r)$ (spherical approximation)	Virial theorem mass estimation
Um,virx(r)	Matches Chandra cluster observations (e.g., M4)	X-ray binary virial trace
Um,mach(?)	Matches Coma radio relic shocks ($M \sim 2-3$)	Merger shock Mach number
Um,merg(t)	$3r_{vir}/(5GM)$	Virial merger crossing time
Um,cross(M)	Matches ~ 2 Gyr for relaxed clusters	Merger timescale (dynamical age)
Um,heat(T)	Balance with duty-cycled AGN	Cool core AGN heating rate
Um,whim(M)	$\mu \sim 0.6$ (mean molecular weight)	WHIM temperature from virialization
Um,cool(T)	$t = E/E$	Cooling time (bremsstrahlung $T > 10^7$ K)
Um,lens(?)	$\theta_E^2 = a \cdot \theta$ Einstein radius	Strong gravitational lensing

10. Additional Specialized Variants

Symbol	F_X Expression	Physical Context
Um,conv(T)	Velocity from $\rho g \, dz \sim v^2$	Convective turnover (mixing length)
Um,lqcf(?)	Friedmann from bounce dynamics	LQC effective Friedmann
Um,lqcp(k)	Power tilt + UV suppression	LQC perturbation pre-bounce
Um,bounc(?)	$\rho \sim 1/(G \cdot t^2)$ singularity capped	LQC bounce density cap
Um,holoent(?)	$\text{Area}(\mathcal{A})/(4G\hbar/c^3)$	Holographic entanglement entropy (Ryu-Takayanagi)
Um,pec(r)	Spherical void: integrate Poisson	Peculiar velocity in voids
Um,voidden(a)	$d \propto a^3$ in matter domination	Void underdensity evolution
Um,jeans(T)	Perturb: $\rho^2 = c_s^2 \cdot k^2 - 4\pi G \rho$	Jeans instability dispersion
Um,fermi(E)	Average $(\rho E/E)^2/2$ per scatter	Fermi-II acceleration (2nd order)
Um,knee(B)	Balance $t_{acc} = t_{age} \sim R/u_s$	Cosmic ray knee from DSA limit
Um,diff(E)	Fit to secondaries (B/C ratio)	Cosmic ray diffusion coefficient

Symbol	F_X Expression	Physical Context
Um,esc(F)	Adjust for penetration K(?)	Photoevaporative escape (exoplanet)
Um,rv(e)	Orbital $v_p = 2pa/P \cdot \sin(i)$	Radial velocity exoplanet detection
Um,upar(r)	$U = G/(n_H \cdot c)$ (ionization parameter)	AGN photoionization parameter
Um,coup(?)	Couple fraction to kinetic energy	AGN feedback thermal coupling
Um,photo(t)	$? \cdot FX \cdot R^3 p / (GM_p/R_p)$	Photoevaporative mass loss rate
Um,puls(t)	Torque $I \cdot dO/dt = -L/O$	Pulsar spin-down magnetic torque
Um,tov(r)	Add GR: P/c^2 energy density, $(1-2GM/rc^2)$	TOV stellar structure GR correction
Um,reljet(z)	Approximate sigmoid for gradual G rise	Relativistic jet velocity profile
Um,after(t)	$N(E) \cdot E^{-p} \times t^{-1.2}$	GRB afterglow synchrotron spectrum
Um,fire(dt)	Internal shocks at $dr \sim c \cdot dt/G^2$	GRB fireball prompt emission
Um,sedov(t)	Integrate momentum: $d/dt(Mv)=0$	Sedov-Taylor blast wave

11. Summary Statistics

Metric	Value
Total named Um variants catalogued	55+
Document source	BBCEquations_04Sept2025.pdf (177 pages)
Total equation pairs (F_UBii + Um) in source	~1,350+ unique positions
Scale coverage	$10^{23} \text{ N} \cdot \text{m}$ (molecular rotors) $? 10^{27} \text{ m}$ (D_universe)
Q_wave std	$6.35 \times 10^4 \text{ J/m}^3$ (47-scale average)

12. References

grok_share_7514fe.txt lines 2676–2762 (first Um catalogue from BBCEquations_04Sept2025.pdf)

grok_share_7514fe.txt lines 6001–6380 (second/final Um catalogue)

PAPER198: FUBii Taxonomy Part 1 (Compact Objects)

PAPER199: FUBii Taxonomy Part 2 (Cosmological/Dark Sector)

PAPER_196: Triadic Master Equation System

13. Operational Implementation Status

The following table classifies each Um variant by its production status in the current codebase (MAIN_1_CoAnQi.cpp, CondensedPhysics.py, CondensedPhysics2.py).

Status	Meaning
■ Operational	Fully implemented, returns numerical result for any input dataset
■ Partial	Core formula implemented; specialized sub-terms use placeholder values
■ Reference	Equation documented for reproducibility; not yet callable from pipeline

Core Um Variants

Symbol	Description	Status	Notes
Um (base)	$\sum_j [\mu_j / r_j \cdot (1 - e^{\{-\gamma t \cdot \cos(\pi t_n)\}}) \cdot \phi_j] \cdot P_{SCm} \cdot E_{react}$	■ Operational	compute_Um_SOURCE4, compute_Um()
Um (Heaviside-amplified)	$Umbase \times (1 + 10^{13} \cdot \Theta(\rho SCm - \rho c)) \times (1 + Aq \cdot \cos(\Delta \omega \cdot t))$	■ Operational	PAPER_421 — integrated v4.75
Um,BZ	Blandford-Znajek power extraction	■ Operational	CondensedPhysics2.py BZ class
Um,haw	Hawking temperature surface gravity	■ Operational	bh_thermodynamics_module.py
Um,qnm	QNM ringdown frequency	■ Operational	CondensedPhysics.py QNM class

Reference-Only Variants (50+)

All remaining Um,X variants catalogued in §2–§10 of this paper are ■ **Reference** status. They contain correct analytical expressions derived from BBCEquations_04Sept2025.pdf but are not yet wired into the main computation pipeline.

Operational upgrade path: To promote any variant from Reference → Operational, implement a calculator class in CondensedPhysics.py following the compute_Um() interface pattern, then register it in the PhysicsTermRegistry.

*Operational status assessed: v4.75 (January 28, 2026 integration).
Um Heaviside amplifier + quasi-periodic modifier now operational (PAPER_421).*